Assignment 3: Derivative of a Gaussian Process

CSDS 611 — Advanced Topics and Methods of Data Science I

Andrei Cozma

Setup

Let $f \sim \mathrm{GP}_{\mathbb{R}}(\mu, C)$, where μ is continuously differentiable and

$$C(x, x') = \sigma^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right), \qquad x, x' \in \mathbb{R}, \quad \sigma^2 \geq 0, \quad \ell > 0.$$

To show that the derivative is a Gaussian process, it is enough to show that its values at any finite collection of inputs are jointly Gaussian. I first construct the derivative as a mean-square limit: a random variable $g(x)$ satisfying

$$\mathbb{E}\left[\left|\frac{f(x+h) - f(x)}{h} - g(x)\right|^2\right] \longrightarrow 0 \qquad (h \to 0). \tag{1}$$

This is the mean-square derivative definition in [1, Section 4.1.1].

Finite differences and their limit

For $h \neq 0$, define the finite difference

$$D_h f(x) = \frac{f(x+h) - f(x)}{h}.$$

It is the slope between the two inputs x and $x + h$. For any N and inputs $x_1, \ldots, x_N$, the vector $(D_h f(x_1), \ldots, D_h f(x_N))^t$ is a linear transformation of the jointly Gaussian values of f at x_i and $x_i + h$. It is therefore jointly Gaussian by the Gaussian transformation property.

Linearity of expectation gives

$$m_h(x) := \mathbb{E}[D_h f(x)] = \frac{\mu(x+h) - \mu(x)}{h} \longrightarrow \mu'(x). \tag{2}$$

For two nonzero step sizes h, r, expanding the covariance gives

$$\begin{aligned} R_{h,r}(x, x') &:= \mathrm{Cov}(D_h f(x), D_r f(x')) \\ &= \frac{C(x+h, x'+r) - C(x+h, x') - C(x, x'+r) + C(x, x')}{hr}. \end{aligned} \tag{3}$$

Write

$$K(x, x') := \frac{\partial^2 C(x, x')}{\partial x \, \partial x'}.$$

The squared exponential kernel has continuous derivatives of all orders. Applying the fundamental theorem of calculus once in each input rewrites (3) as

$$R_{h,r}(x, x') = \int_0^1 \int_0^1 K(x + th, x' + sr) \, ds \, dt \longrightarrow K(x, x') \qquad (h, r \to 0). \tag{4}$$

For a fixed x, the identity $\mathbb{E}[Z^2] = \mathrm{Var}(Z) + (\mathbb{E}[Z])^2$ gives

$$\begin{aligned} \mathbb{E}\big[|D_h f(x) - D_r f(x)|^2\big] &= R_{h,h}(x, x) + R_{r,r}(x, x) - 2R_{h,r}(x, x) \\ &\quad + \big(m_h(x) - m_r(x)\big)^2 \longrightarrow 0. \end{aligned}$$

Thus the difference quotients are Cauchy in mean square. Completeness of the space of square-integrable random variables gives a limit $g(x)$ satisfying (1).

Gaussianity and moments of the derivative

Mean-square convergence preserves expectations and covariances, so (2) and (4) imply

$$\mathbb{E}[g(x)] = \mu'(x), \qquad \mathrm{Cov}(g(x), g(x')) = K(x, x'). \tag{5}$$

For any finite collection $x_1, \ldots, x_N$, the Gaussian vectors of finite differences converge in mean square, hence in probability, to $(g(x_1), \ldots, g(x_N))^t$. A limit in probability of Gaussian vectors is Gaussian: their mean vectors and covariance matrices converge here to those in (5), and these parameter limits determine the limiting Gaussian distribution. Since N and the inputs were arbitrary, g is a Gaussian process.

To interpret g as an ordinary derivative, I use the sample-function regularity of the squared exponential kernel: the centered process $f - \mu$ has a version with continuously differentiable sample functions [2, Section 2]. Here a version agrees with the original process almost surely at each fixed input. Adding the given continuously differentiable μ preserves this regularity. For this version, the difference quotients converge almost surely to the ordinary derivative. They also converge in mean square to $g(x)$, so uniqueness of limits in probability gives $f'(x) = g(x)$ almost surely at each fixed x.

Covariance kernel and conclusion

Differentiating the kernel once in each input gives

$$\frac{\partial C(x,x')}{\partial x} = -\frac{x-x'}{\ell^2}C(x,x'), \qquad \frac{\partial C(x,x')}{\partial x'} = \frac{x-x'}{\ell^2}C(x,x').$$

Using the product rule,

$$\begin{aligned} K(x,x') &= \frac{\partial}{\partial x'}\left[-\frac{x-x'}{\ell^2}C(x,x')\right] \\ &= \left(\frac{1}{\ell^2} - \frac{(x-x')^2}{\ell^4}\right)C(x,x'). \end{aligned}$$

This is symmetric in its two inputs and gives $K(x,x) = \sigma^2/\ell^2 \geq 0$. It is a valid covariance kernel because it is the covariance of the process g constructed above.

Therefore, with the ordinary derivative taken for the differentiable version,

$$\boxed{\begin{gathered} f' \sim \mathrm{GP}_{\mathbb{R}}(\mu', C_\partial), \\ C_\partial(x,x') = \sigma^2\left(\frac{1}{\ell^2} - \frac{(x-x')^2}{\ell^4}\right)\exp\left(-\frac{(x-x')^2}{2\ell^2}\right). \end{gathered}}$$

References

[1] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006, Section 4.1.1. Chapter 4.

[2] C. J. Paciorek and M. J. Schervish. *Nonstationary Covariance Functions for Gaussian Process Regression*, Section 2. Paper.